Case Study 2: Technical Presentation

STAT 946 – Case Studies in Data Science

Dario Greco Lovera Lokeswara Sanika Poojary Sophia Feng

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Section 1

Intro

Background, stakeholders, operational value

Organization: Toronto Transit Commission (TTC)
Primary stakeholder: Director of Station Operations and Maintenance

Operational context:

- Subway stations rely on HVAC systems that draw outside air
- During outdoor $PM_{2.5}$ spikes, filters degrade faster and indoor air quality risk increases
- Forecasts enable proactive station ventilation actions (pre-ventilate, reduce intake during spikes, recirculation)

Research questions

Primary:

How accurately can hourly $PM_{2.5}$ in Ontario be forecasted at short horizon ($h = 4$ hours) using meteorological and environmental data?

Case Study 1 vs Case Study 2

Limitations identified in case study 1:

- Traffic data only available daily, forcing hourly $PM_{2.5}$ to be averaged into daily values, losing spike timing
- Traffic impact was weak. Weather and seasonality were the true drivers

Key changes in case study 2:

- Remove unhelpful traffic signal, focus entirely on weather drivers
- Improve data resolution by using hourly weather and historical $PM_{2.5}$ to predict future hourly $PM_{2.5}$
- Focus on prediction rather than interpretability
- Focus on the closest air quality and meteorological station

Section 2

Data and preprocessing

Data Sources and Cleaning

Air quality: hourly pollutant $PM_{2.5}$ concentrations in downtown Toronto

Weather: hourly meteorology (temperature, humidity, wind speed, precipitation, air pressure, dew point temperature) in downtown Toronto

Alignment:

- Join by timestamp (date-hour) at a Toronto monitoring station
- Hourly data from 2019/06/22 to 2025/12/31 **(total 56,362 rows)**

Core steps:

- Merge: hourly weather with hourly $PM_{2.5}$ by timestamp
- Missingness handling:
  - $\sim 1\%$ of $PM_{2.5}$ data was missing
  - $PM_{2.5}$ missingness was forward filled up to 4 hours, else the remaining rows were dropped
  - Weather variables all had $< 1\%$ missingness – missing variables were forward filled
- Feature construction:
  - Cyclical time features for hour and month

Section 3

Baseline models

1. Establishing the Baseline

The Forecasting Objective: Predict the **+4h future PM2.5 concentration** utilizing a rolling **24-hour historical lookback window**.

The Challenge: Feeding 24 hours of multiple weather variables (temp, wind, humidity, etc.) into a linear model creates a highly dimensional, collinear feature space.

The Solution: Lasso (L_1 Regularization)

$$\hat{\beta} = \arg\min_{\beta} \underbrace{\sum_{t} (\text{PM2.5}_{t+4} - \beta_0 - \beta^\top X_{t-23:t})^2}_{\text{1. Minimize Error}} + \underbrace{\lambda \|\beta\|_1}_{\text{2. Penalty Term}}$$

Why it fits our data:

- **Feature Selection:** The penalty shrinks the weights of irrelevant lags and weather noise to *exactly zero*.

2. Implementation Workflow

1. **Data Transformation**
   - **Flattening:** Reshaped the 24h 2D historical weather windows into 1D vectors.
   - **Scaling:** Standardized (Z-scores) to ensure unbiased penalty application across varying physical units.
2. **Strict Chronological Split & Time-Series CV**
   - **The Split:** Time-series data is never randomly shuffled. We allocated the **first 85% of data for Training & CV**, keeping the **final 15% completely isolated for Testing**.
   - **Expanding Window (5 Folds):** Standard cross-validation leaks future data into the past. We used `TimeSeriesSplit` on the 85% block to both train on a growing history and validate on the immediate rolling future.
3. **Hyperparameter Fine-Tuning**
   - **Alpha Tuning:** Utilized `LassoCV` (`max_iter=10000`) to dynamically sweep for the optimal penalty strength ($\alpha \equiv \lambda$) across our 5 chronological folds.

3. Limitations & The Path Forward

What Lasso Did Well:

- Filtered out noise and provided a solid RMSE benchmark.

The Limitation:

- Lasso is fundamentally a **linear** model constrained by the 24h window's dilution effect.
- It struggles to capture **non-linear, dynamic shocks**.

The Solution: N-HiTS

- We need a complex architecture capable of separating slow atmospheric trends from sudden weather shocks.

Section 4

Complex model: N-HiTS

Why N-HiTS

Definition

N-HiTS: Neural Hierarchical Interpolation for Time Series

Goal: capture nonlinearities and multi-scale temporal structure with a deep, compute-heavy model.

Why it fits air-quality data:

- multi-resolution learning (slow trends + fast spikes)
- non-linear mapping from weather and seasonality to $PM_{2.5}$
- residual learning to decompose signal into components

Input and output shapes:

$$x_t \in \mathbb{R}^{L \times d}, \quad d = 11$$

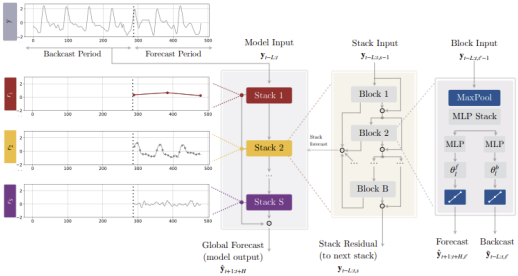
$$\hat{y}_{t+1:t+24} = f_\theta(x_t), \qquad \hat{y}_{t+4} = [\hat{y}_{t+1:t+24}]_4$$

- lookback: $L = \{24\}$
- horizon head: 24-step output, but we report the 4-hour-ahead value

N-HiTS Architecture: Hierarchical multi-resolution

The N-HITS model as "3 frequency stacks":

- Stack 1 (macro) → Block 1: max-pool ×4, $n_\theta = 6$
- Stack 2 (mid) → Block 2: max-pool ×2, $n_\theta = 12$
- Stack 3 (fine) → Block 3: no pooling, $n_\theta = 24$



Our N-HiTS structure

Three frequency blocks:

Block	Pool size	Resolution captured	Forecast knots
Block 1	4	long-term trends	6
Block 2	2	medium variation	12
Block 3	1	short-term spikes	24

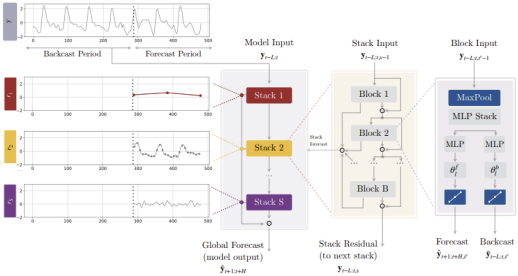
Forecast combination:

- each block produces a forecast
- forecasts are summed
- we evaluate the **4-hour-ahead prediction**

How one N-HiTS block works

Pipeline inside each block:

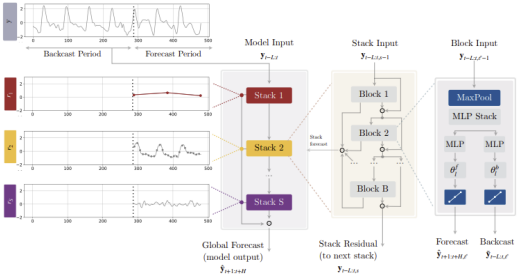
1. **Pooling (multi-scale view)**
   Compress the input window to capture patterns at a specific time scale.
   - Block 1: coarse trends (pool $\times 4$)
   - Block 2: medium variation (pool $\times 2$)
   - Block 3: fine detail (pool $\times 1$)
2. **MLP encoder**
   A small neural network learns a hidden representation of the time window.
3. **Backcast**
   The block reconstructs the part of the input signal it can explain.



How one N-HiTS block works

Pipeline inside each block:

4. **Residual update**
   The explained signal is removed so the next block focuses on what remains.
5. **Forecast generation**
   The block predicts a small set of forecast knots.
6. **Interpolation to horizon**
   These knots are linearly interpolated to produce a 24-step forecast.



Final prediction:

$$\hat{y} = \hat{y}^{(1)} + \hat{y}^{(2)} + \hat{y}^{(3)}$$

Each block contributes a forecast at a different temporal resolution.

Target-specific residual learning

Split features:

- target: $PM_{2.5}$
- exogenous: weather + time features

Residuals are applied only to the target channel:

$$r^{(1)} = y_{\text{target}} - \widehat{x}^{(1)}_{\text{target}}$$

For block 2 and 3, the input is:

$$x^{(k)} = \left[r^{(k-1)},\ \text{exog}\right]$$

Final forecast is the sum of all block forecasts:

$$\hat{y}_{t+1:t+24} = \widehat{y}^{(1)} + \widehat{y}^{(2)} + \widehat{y}^{(3)}, \qquad \hat{y}_{t+4} = \left[\hat{y}_{t+1:t+24}\right]_4$$

Training and evaluation

Data:

- standardized 11 features ($PM_{2.5}$ + weather + seasonal sin/cos)
- split: 70% train, 15% validation, 15% test (time-ordered)

Optimization:

- loss: MSE
- optimizer: Adam
- epochs: up to 50
- early stopping: patience = 8 (best val RMSE checkpoint)

Metrics:

- RMSE on test set for 4-hour-ahead prediction

Compute:

- GPU (Waterloo Remote Server), batch size 128

Hyperparameter tuning

We run a small hyperparameter sweep centered on best-performing configs:

- lookback $L \in \{24, 48\}$
- hidden width $\in \{128, 256, 512\}$
- learning rate $\in \{0.001, 0.0005, 0.0001, 0.00005\}$

Selection rule:

- choose lowest validation RMSE
- trainable parameters: 1,453,890 parameters

Chosen config

$L = 24$, hidden width $= 512$, learning rate $= 0.00005$

Section 5

Comparison and evaluation

Model Performance - Forecast

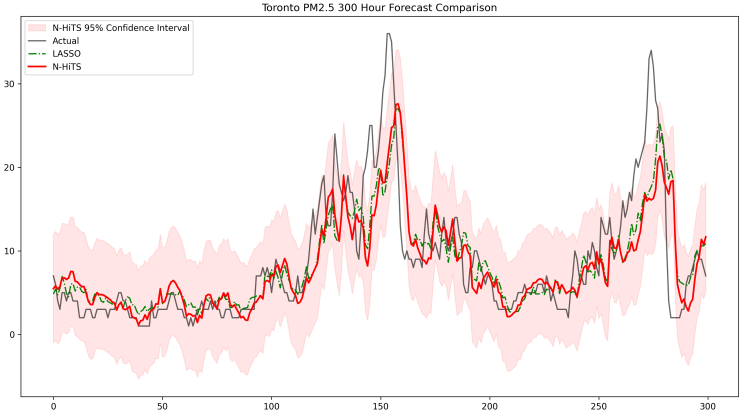


Figure 1: Toronto PM2.5 4-Hour Forecast Model Comparison

Model Performance - RMSE Comparison

Table 2: Model RMSE Comparison

Model	RMSE	MAE
LASSO	6.7789	4.1672
N-HiTS	6.8246	4.1761

- The LASSO and N-HiTS models produce very similar RMSE and MAE values.
- The N-HiTS model's ability to capture non-linearities and multi-scale patterns does not translate into a significant improvement in forecasting accuracy for this dataset, suggesting that the underlying dynamics may be relatively simple.

Overall Forecasting Accuracy and Effectiveness

- Typical hourly $PM_{2.5}$ levels in Ontario often range from ~5–30 μg/m^3, with occasional spikes above 50 μg/m^3 during pollution events. An RMSE of ~6.7 μg/m^3 therefore corresponds to an average prediction error of roughly 20–30% of typical concentration levels.
- This level of error is reasonable for short-term environmental forecasting, where pollution levels are highly variable and influenced by complex atmospheric processes.
- This forecasting system should be treated as an early-warning decision support tool, not a precise measurement system.
- Its value lies in anticipating pollution spikes early enough for operational adjustments, rather than predicting exact $PM_{2.5}$ concentrations.